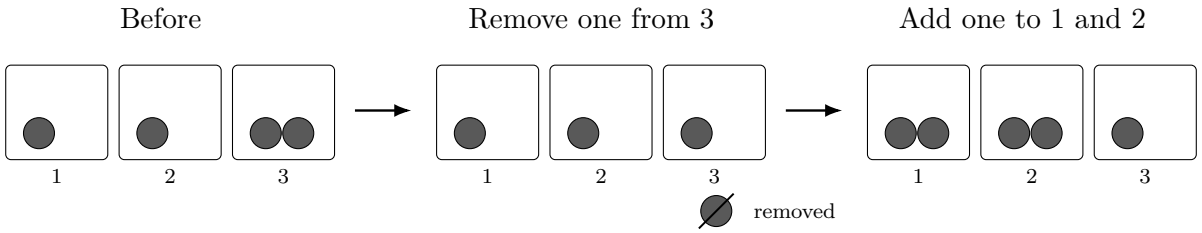
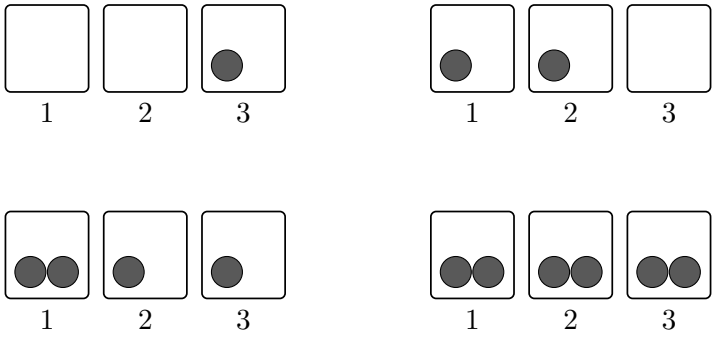
Take turns. Remove one counter from a nonempty bucket. You may then add counters only to buckets strictly to its left. If you cannot move, you lose.

For Problems 1–3, add at most one counter to each bucket to the left of the one you took from.



Problem 1: Play each start with a partner. Which starts let the first player win whatever the second player does?

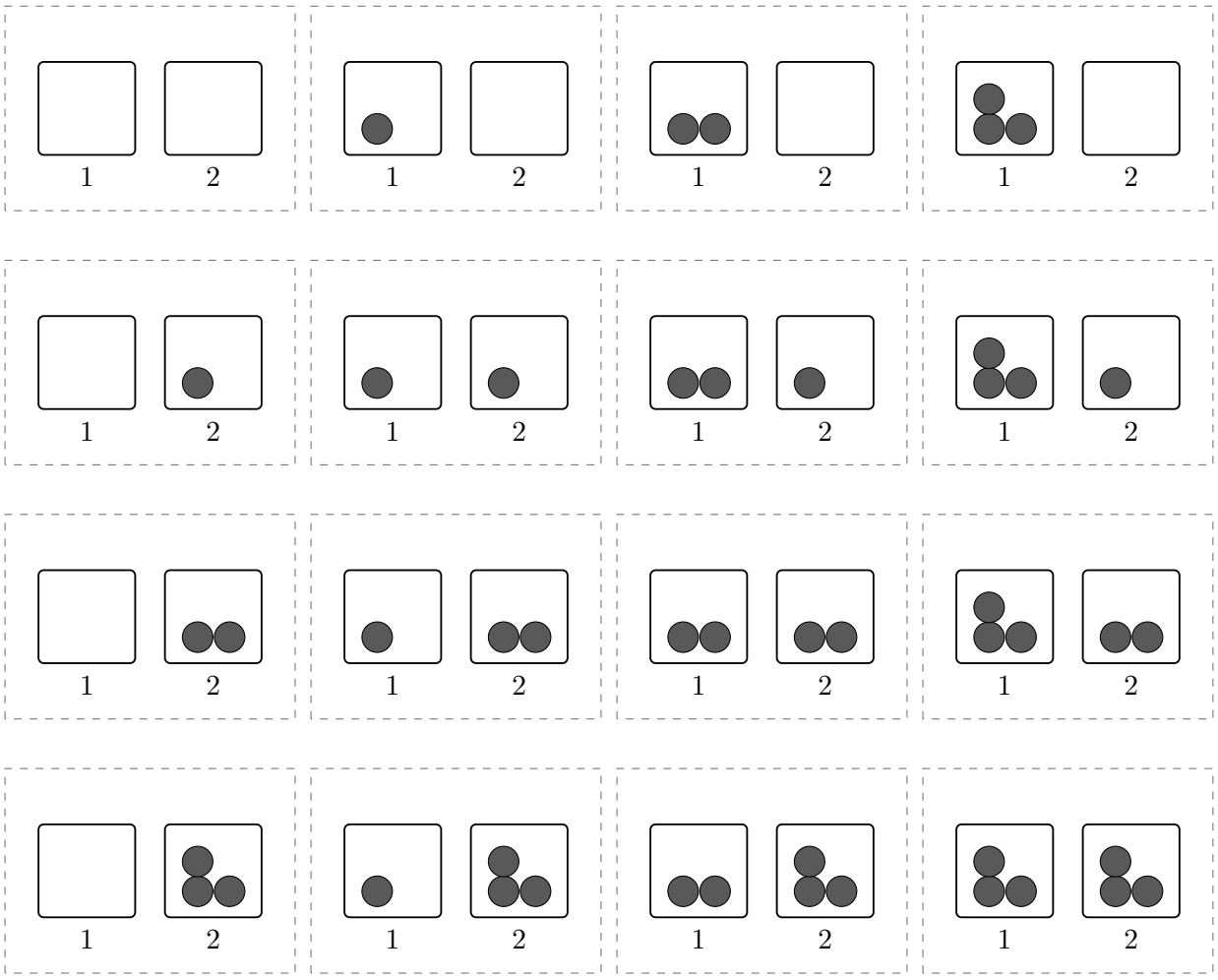


Bucket 1 | Bucket 2 | Bucket 3

Supply

Removed

Problem 2: Use two buckets. Sort these starts into two groups: the first player can force a win, or the second player can force a win.



Problem 3: Play from these three-bucket starts. Can you find a rule for choosing a winning move whenever there is one?

